

Output:

```
"C:\Program Files\Java\jdk-24\bin\java.exe" --enable-  
Future Value after 5 years = 16105.10  
Process finished with exit code 0
```

Analysis:

Time Complexity: $O(n)$

Optimization:

Problem

For large values of n , recursion creates many function calls and uses extra memory.

```
23     public static double futureValueIterative(double pv,  
24                                           double rate,  
25                                           int years) {  
26         for(int i = 0; i < years; i++) {  
27             pv *= (1 + rate);  
28         }  
29         return pv;  
30     }
```